Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Adagard
b. Team leader name: Nivakaran Shanmugabavan
c. Problem Statement: Challenge 1 (BFSI) · Stopping the payment a scam victim approves (APP fraud)

Brief about the idea

FirstStep is the last checkpoint before the money leaves. It catches the scam at the moment of payment, explains it to the customer in Sinhala, Tamil or English, and gives the bank and regulator a record that can't be quietly changed.

In an authorised push payment scam the customer passes every check herself: right login, right OTP, her own decision to pay. Regulators are moving that loss onto banks.

Oct 2024

UK: mandatory reimbursement of APP scam losses, up to GBP 85,000 per claim

Dec 2024

Singapore: Shared Responsibility Framework puts duties on banks and telcos

Jun 2026

Sri Lanka: CBSL makes “Be Scam Proof” its consumer-protection theme

1 Pay a new payee

Rs 30,000 to a new payee: before any code, the bank asks what it is for

2 Scam check

She pastes the WhatsApp offer: STOP, and why, at the moment of payment

3 Explained

In Sinhala, with every figure checked against the tools

4 Sealed record

Evidence for disputes, reimbursement and the regulator

Also: USD 88–114 bn lost to online scams across Asia-Pacific in 2025 (UNODC); only 22% of victims recover anything (GASA).

Opportunities

Why not existing apps?

- Google Messages and ScamShield protect the phone and need the victim to doubt the offer: 53% of victims aged 65+ who didn't check thought it unnecessary (CommBank, 2026)
- FirstStep protects the payment: it acts at the transfer, on any phone, whatever the channel (WhatsApp, calls)
- A bank-side control, so it fits Singapore's and the UK's duties on banks

How it solves the problem

- At a new payee the bank pauses and asks what the payment is for, the moment regulators single out as high risk
- Plain-language reasons in the customer's language, where generic English warnings fail
- Policy decides, Gemini reads: the bank can tell a regulator exactly why a payment was stopped
- A sealed, verifiable record for disputes and reimbursement claims

USP

- Protects the payment, not just the phone
- Live and tested on Cloud Run: 11/11 end-to-end checks
- Built for Sri Lanka first, APAC-ready: Sinhala, Tamil, LKR and CSE today; a new market is mostly data (regulator register, scam corpus, OKF pattern pack, Gemini language) plus a new policy version, no new services
- Next markets: Singapore (bank duties under the Shared Responsibility Framework); Malaysia and India (Tamil already supported)

List of features offered by the solution

1 Payment checkpoint

A new payee or unusual behaviour pauses the transfer: the bank asks what it is for, checks the answer with Scam Shield, and only then sends a one-time code.

2 Scam Shield

Paste a WhatsApp, SMS or email offer. Gemini reads the red flags, policy SHIELD-P001 decides: a STOP / CHECK / CLEAR stamp with what to do next.

3 Explained in her language

Ask why, by text or voice, in Sinhala, Tamil or English. Gemini answers from the platform's tools; any unchecked figure is withheld.

4 Sealed evidence

Each decision is hashed with its model and policy and anchored in a contract; change one field and verification fails. Readable over OData.

5 Scam knowledge, open

Scam patterns in Google's Open Knowledge Format, with sources and trust tiers, cited in every verdict.

6 Slowing down is cheap

Scammers sell urgency. The cash planner shows what rushing a sale costs, and the cheapest way to raise money calmly.

Process flow diagram

Transfer
Kamala pays Rs 30,000 to a new payee

Gateway
Behavioural and payee-network risk score (heuristic)

Pause & ask
“What is this payment for?” She pastes the WhatsApp offer

Scam check
Gemini reads the red flags; SHIELD-P001 decides

STOP
Cancel is the default; continuing needs consent, then the code

Then the customer's question and the bank's record

Ask why
Assistant answers in Sinhala; invented figures are withheld

Decide
She cancels, or acknowledges the risk and confirms with a code

Sealed
Verdict and transfer decision hashed with model and policy

Verify
Any change later fails against the contract

Report
Bank and regulator read every decision over OData

The step-up runs Scam Shield on the customer's answer before any code is sent; feeding the verdict into the gateway's risk score is next.

Design: the verdict is the product

One decision, one stamp.

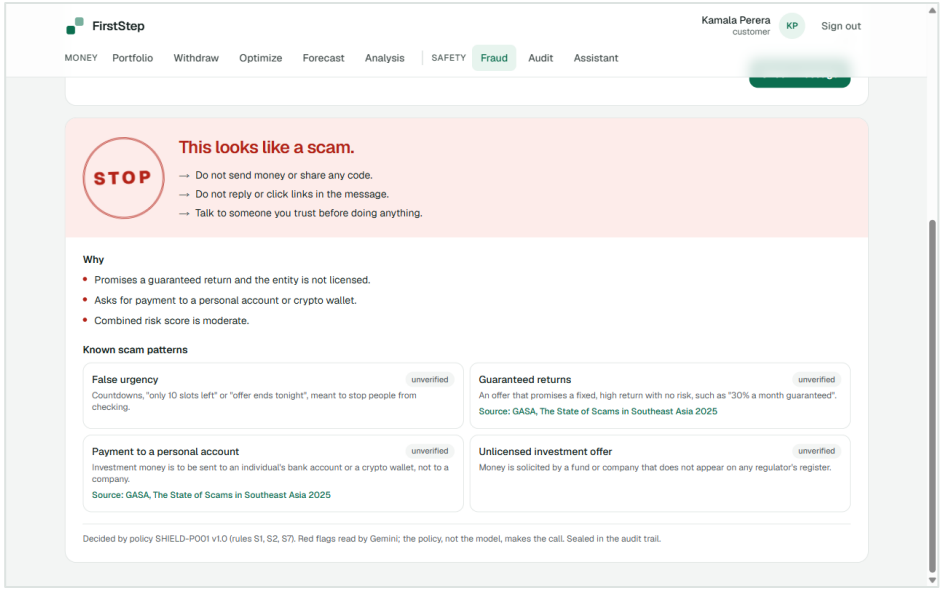
The visual world is a bank passbook: paper, ink, savings-book green for safe, stamp red for stop. The verdict is a stamp with a word and a shape, never colour alone, then what to do, then why.

STOP This looks like a scam

CHECK Some warning signs

CLEAR No warning signs found

UNSURE Too little to call it safe



Live screen. Every text colour clears 4.5:1 contrast in light and dark; layouts are checked at phone width (390 px).

Architecture diagram of the proposed solution

Customers
Web app and mobile app

Banks, regulators
OData v4 clients (Excel, BI)

Google Cloud Run · each service verifies every request itself

Web (Next.js)
Scam Shield, planner, audit, assistant UI

Platform
Accounts, roles, portfolios, token signing

C2 Fraud & Scam
Gemini red flags, SHIELD-P001, gateway G1-G7, step-up, OKF bundle

C1 Portfolio
Fuzzy GA cash planner, MOEA/D, forecast bands, statement import

C4 Assistant
LangGraph agent on Gemini, grounding check, Trust Panel

C3 Audit
Hash-committed decision log; AuditRegistry.sol on the audit service chain

Gemini
Red flags, agent reasoning, Sinhala and Tamil answers

Secret Manager
Signing key, service-client secrets

Cloud Build
Images per service, kept in Artifact Registry

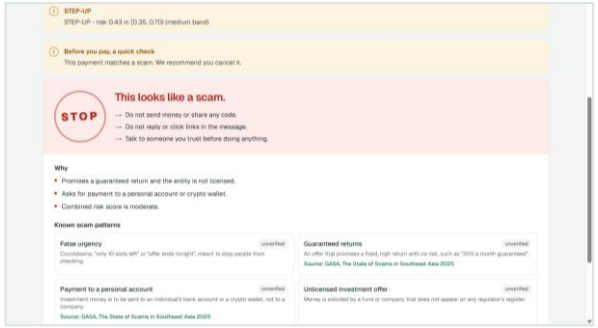
GitHub Actions + WIF
Keyless deploy on every push

Technologies used in the solution

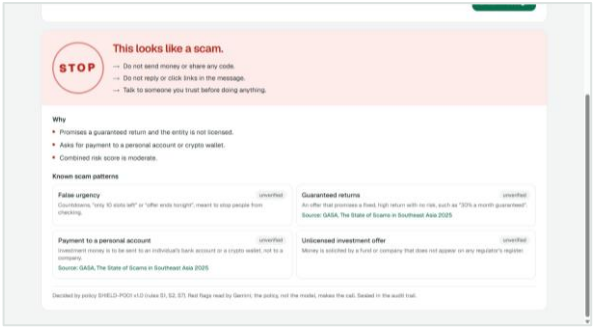
Live	Gemini (red flags, agent, Sinhala/Tamil) · Cloud Run (6 services) · Cloud Build · Artifact Registry · Secret Manager · Workload Identity Federation · Open Knowledge Format (OKF) · Cloud Logging
Live	Next.js 16 + React 19 · FastAPI · Express · LangGraph · Expo (React Native) · Hardhat EVM + Solidity · OData v4 · RS256 / JWKS
Offline	TimesFM fine-tuned with LoRA on a Colab T4 + LSTM residual head; walk-forward evaluation
Planned	Vertex AI, ADK + Agent Engine, BigQuery ML AI.FORECAST, Document AI, Vertex AI Search, Pub/Sub, Firestore, Firebase Auth, Cloud Armor, VPC-SC, Blockchain Node Engine

Status as deployed on 18 Oct 2026. The README's technology map lists the same split.

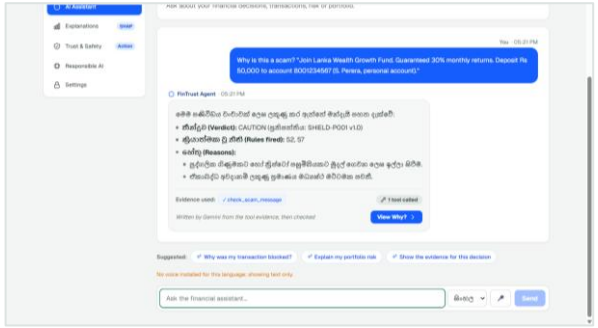
Snapshots of the prototype



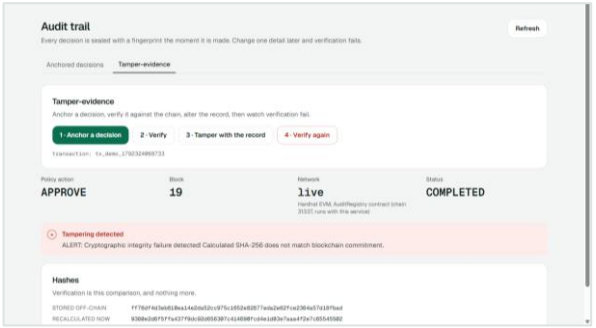
Payment checkpoint: the bank asks what the payment is for and shows STOP before any code



Scam Shield: STOP stamp, reasons, cited OKF scam patterns; red flags read by Gemini

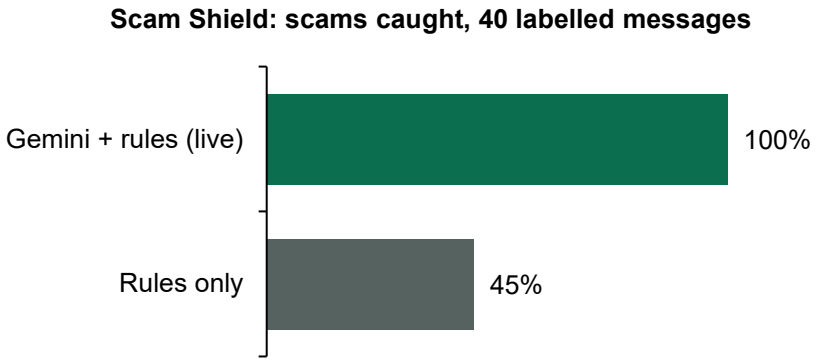


Assistant answering in Sinhala, from the scam-check tool



Audit: record altered, verification fails against the contract

Prototype performance report / benchmarking



Synthetic test set (40 msgs): recall 45% → 100% with Gemini, 0 false positives

Gemini reading red flags lifts recall from 45% to 100% at the same zero false-positive rate (precision 1.00). Honest limits: 40 synthetic English messages written by the team (eval/scam_synthetic_v1.jsonl); Sinhala and Tamil not yet measured; a real labelled Sri Lankan set is next.

Forecasting (offline): fine-tuned TimesFM hybrid, −6.8% pinball loss vs LSTM.

11 / 11 live end-to-end checks pass on Cloud Run

176 + automated tests passing: platform 63, fraud 46, assistant 22, web 38, mobile 7, plus the audit suite

401 / 403 no token or wrong role is refused, by every service

6 / 6 services live on Cloud Run, redeployed by CI on every push

Additional details / future development

Already in place

Honest by design

The README separates what is live, offline and planned; negative results are reported.

Responsible AI

Policies decide, figures are grounded, privacy and advice checks run on every answer; family guardian alerts wait for the user's consent.

Next

Verdict into the score	Feed the scam verdict into the gateway's risk score, not only the step-up.
Real-world eval	Labelled Sri Lankan messages, Sinhala and Tamil, with partner banks.
Gemini on Vertex AI + ADK	Move the agent to Agent Engine; Document AI for statements.
Public anchoring	Anchor the log on a public testnet via Blockchain Node Engine.
WhatsApp bot	Forward a message, get the verdict back in the same chat.

Provide links to your:

- **GitHub Public Repository:** https://github.com/Adagard-Trios/AI-Buildathon-2026
- **Demo Video Link (3 Minutes):** ADD THE VIDEO LINK
- **Final Product Link:** https://firststep-web-214625202650.us-central1.run.app

Try it

Register any email: every account starts with a demo portfolio of CSE, US and crypto holdings.

Fraud → Check a transfer: tick New beneficiary, amount 30,000 → Score → answer “what is it for?” by pasting an offer → STOP → cancel, or tick the warning and enter the code. (If it says BLOCK, type a new name in User.)

Fraud → Scam Shield: paste any offer on its own and see the same verdict, reasons and scam patterns.

Audit → Tamper-evidence: anchor, verify, tamper, verify again.

Services scale to zero to keep cost near nothing: the first click on a quiet service takes ~15 s.

Built to be trusted

1 Verify every request

Each service checks RS256 tokens against the platform's JWKS; no token means 401, wrong audience or role 403.

2 Least privilege

Roles come from deployment config only; staff addresses cannot be self-registered.

3 Your data only

Row-level reads: customers see only their own records; auditors read all, alter nothing.

4 Fresh proof for risk

STEP-UP one-time codes on risky transfers, even inside a valid session.

5 Fail closed

Misconfiguration refuses instead of allowing; denials are sealed in the audit trail.

6 No invented numbers

An answer with a figure no tool produced is withheld.

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Thank you

